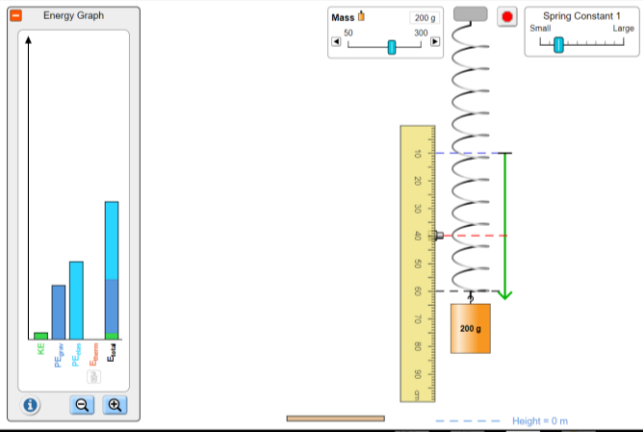
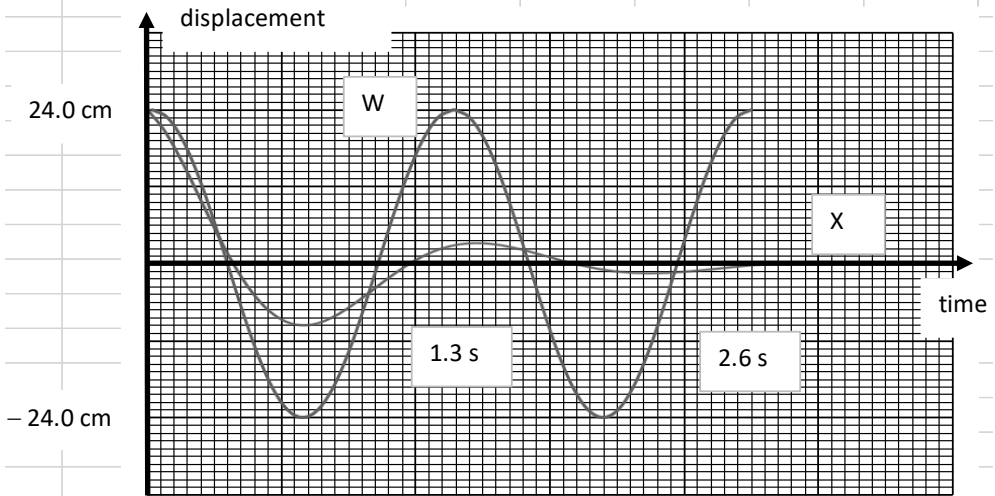


3	(a)	(i)	$\omega = 2\pi f \quad [\text{M1}]$ $\rightarrow f = \frac{1}{2\pi} \sqrt{\frac{10.5}{0.450}} = 0.76879 \sim 0.77 \text{ Hz} \quad [\text{A1}]$
		(ii)	From 50 cm to 80 cm position, a change of 0.300 m: $\Delta GPE = Mg(\text{extension}) = 0.450 \times 9.81 \times 0.300 = 1.32435 \text{ J} \quad [\text{M1}]$ When mass was initially held at 50 cm position, spring was already experiencing EPE, as unstretched length of spring is 32 (or 32.5 cm), therefore initial EPE is due to $e = 18 \text{ cm}$. When mass ends up at 80 cm position, spring would have stretched by $e = 48 \text{ cm}$. $\Delta EPE = \frac{1}{2}k(e_2^2 - e_1^2) = \frac{1}{2}(10.5)(0.480^2 - 0.180^2) = 1.0395 \text{ J} \quad [\text{M1}]$ So, at the position of 80 cm, the KE present in the mass would be: $\Delta KE = 1.32435 - 1.0395 = 0.28485 \text{ J} \quad [\text{C1}]$ $0.28485 = \frac{1}{2}mv^2 \rightarrow v = 1.12517 \sim 1.13 \text{ m s}^{-1} \quad [\text{A1}]$
			

		When considering energy conversion for a vertical spring-mass system, there are GPE, EPE and KE. This is at a position close to the maximum displacement.
(b)	(i), (ii)	 Fig. 3.3 Labels – <u>Displacement</u> of +/- 24.0 cm, 2 cycles of <u>period</u> 1.3 s (for Line W) [B1] When spring-mass is at equilibrium, reading off the graph, the length of the spring should be 74.0 cm. Given the start length of the spring is 50.0 cm, therefore the displacement of free oscillations will be 24.0 cm. Line W – Accept +ve/-ve <u>Cosine</u> graph [B1] Line X – Check-by-eye <u>decreasing displacement</u> over time, 2 cycles of <u>consistent larger period</u> than Line W [B1]

4	(a)	Interference is the superposing of two or more waves to produce	
---	-----	---	--